Classwork

Uniform Centripetal Motion & Universal Gravitation

Class: ____________________ **Date:** _______________

Part I: Uniform Centripetal Motion

Problem 1: A red ball moves in a **horizontal** circle at constant speed.

(a) Draw velocity, acceleration, and net force at three different points on the circle.

(b) Which vector(s) are tangent to the circle? Which vector(s) point toward the center?

(c) Explain why the object can accelerate even though the speed is constant.

Problem 2: A green car drives around a flat **horizontal** circular track at constant speed.

(a) Draw a sketch of what is going on (velocity, acceleration, centripetal force).

(b) Draw a Free Body Diagram (FBD) (include F_g and F_N). (Hint: The y forces cancel anyway)

(c) Identify the force that provides the centripetal force.

(d) Predict the car's path if that force suddenly disappears. Describe in 1-2 sentences. In the extra space, feel free to draw it to visualize this.

Problem 3: A blue car of mass 1000 kg travels in a circular loop with a radius of 2 km at a constant speed of 55 km/h. (Hint: Be careful with units).

(a) Draw a sketch of what is going on (velocity, acceleration, centripetal force).

(b) Draw a Free Body Diagram (FBD) (Include F_g and F_N).

(c) What is the sum of the y forces?

(d) What is the sum of the x forces?

(e) What is the centripetal acceleration of the car?

(f) What is the centripetal force of the car?

(g) What type of force is in reality playing the role of centripetal force in this scenario?

(h) Suppose the car goes at double the speed. How much would the acceleration value increase/decrease by? (Find the factor without using a calculator.

(i) Suppose the radius is increased by a factor of 10. How much would the acceleration value increase/decrease by? (Find the factor without using a calculator.)

(j) Suppose the car's mass is tripled. How much would the force increase/decrease by? (Find the factor without using a calculator.)

Problem 4: A purple car and a yellow car are racing each other in a circular loop. Both cars start with the same speed and with the same radius. The only difference is Car A has twice the mass of Car B.

(a) Which car requires a larger centripetal force? Explain.

(b) Which car has a larger centripetal acceleration? Explain.

(c) If we wanted both cars to exert the same force, which car should have a radius change? What is the factor by which the radius should change?

(d) If we wanted both cars to exert the same force, which car should have a velocity change? What is the factor by which the velocity should change?

Problem 5: A brown roller-coaster car of mass 500 kg moves through a **vertical** circular loop of radius 12 m at a constant speed.

(a) At the **bottom** of the loop:

i. Draw a sketch of where the car is.

ii. Draw a Free Body Diagram (FBD) and clearly label all forces.

iii. Which forces now point toward the center of the loop? (up, down, left, or right)

(b) Write the force equation to find the total normal force in the radial direction at the bottom of the loop. (Choose inward as positive.)

(c) Solve for the total normal force.

(d) Now consider the **top** of the loop:

   i. Draw a sketch of where the car is.

   ii. Draw a Free Body Diagram (FBD) and clearly label all forces.

   iii. Which forces now point toward the center of the loop? (up, down, left, or right)

(e) Write the force equation to find the total normal force in the radial direction at the bottom of the loop. (Choose inward as positive.)

(f) Solve for the total normal force.

(g) Compare your expressions for the normal force at the top and bottom. Where is the normal force larger? Why? Where is the normal force smaller? Why?

Part II: Universal Law of Gravitation

Problem 6: A 100 kg turquoise satellite orbits Earth in a circular path.

(a) Identify the force that keeps the satellite in orbit.

(b) According to Newton's Third Law:

i. Does Earth exert a force on the satellite?

ii. Does the satellite exert a force on Earth?

(c) Calculate F_g in both ways. (Hint: When $F_g = mg$ and the other equation).

(d) Compare the magnitudes of these two forces. Are they the same or different?

(e) Using $a = \frac{F}{m}$, find the acceleration of the Earth & the acceleration of the satellite.

(f) Explain why the satellite accelerates noticeably but Earth does not.

Problem 7: A gray satellite of mass 1500 kg orbits Mars at an altitude of 650 km above the surface. The satellite moves at a constant speed of 1750 km/h. ($M_{mars} = 6.42 \times 10^{23}$ kg) (Hint: Be careful with units).

(a) Determine the **total orbital radius** of the satellite. (Hint: radius of the planet as well as the altitude.)

(b) Convert the satellite's speed into **meters per second**.

(c) Using your results from parts (a) and (b), calculate the centripetal acceleration of the satellite.

(d) Identify the force that provides this centripetal acceleration.

(e) Explain why this force is able to keep the satellite in circular motion rather than pulling it straight down.

Problem 8: Two satellites, A and B, orbit Earth in the **same circular orbit**. Satellite A completes one orbit in 90 min, while Satellite B completes one orbit in 120 min.

(a) Which satellite has the greater orbital speed? Explain how you know without using a calculator.

(b) Orbital frequency f is defined as the number of orbits completed per unit time.

i) Write a very simple expression for relating 1 orbit to some number of time (period) T.

ii) Calculate the orbital frequency of Satellite A (in orbits per minute).

iii) Calculate the orbital frequency of Satellite B (in orbits per minute).

(c) Because the satellites move at different rates, one gains angular position on the other.

i) Determine how many orbits Satellite A completes in one minute. Hint (Think about part b)

ii) Determine how many orbits Satellite B completes in one minute. Hint (Think about part b)

iii) Find how much of an orbit Satellite A gains on Satellite B in one minute.

(d) When Satellite A has gained one full orbit relative to Satellite B, it will pass directly over Satellite B again. Using your result from part (c), calculate how long it takes for Satellite A to pass directly over Satellite B again.